Real-Time Sui

GraphQL Subscriptions and the Data Stack After JSON-RPC

01 / CHOOSE

Map the data stack

02 / BUILD

A feed that survives a drop

03 / MIGRATE

One call in your own app

A payment screen is the worked example for the whole lesson.

The Sui stack, after Basecamp.

LAYER	WHAT IS THERE NOW	STATUS
Apps + agents	Samsung Wallet USDC (82M phones) · Slush · DeepBook App API	Live · Samsung late Oct
Agent payments	Agent Payments (x402, MPP, Alibaba Cloud) · Payment Kit · Agent Pass	Testnet · Pass announced
Accounts + money	Passkeys + aliases · allowances · address balances · USDC + CCTP V2	Live · allowances testnet
Throughput	Programmable Tunnels: 40.6M TPS offchain, settled to mainnet	Live · no SDK yet
Data ← TODAY	gRPC · GraphQL + subscriptions · archive · indexer. JSON-RPC off	Live
Storage + privacy	Walrus · Walrus Console (API + MCP) · Seal encryption	Live
Infra + tools	AWS · Google Cloud · Alibaba Cloud nodes · Sui skills + docs MCP	Live

Day 1 and Day 2 built the accounts + money row. Today: the data row every other row reads from.

What Basecamp changes for your data layer.

AGENTS ARE CLIENTS

Agents read and pay per call

x402 + MPP. They need live feeds, not polling.

PHONE-SCALE PAYMENTS

82M wallets, gasless USDC

Receipts must survive drops: cursors.

MONEY WITHOUT COINS

Watch balance changes

Address balances and allowances skip coin objects.

TUNNELS

40.6M TPS stayed offchain

Your feed sees settlements, not each message.

DATA OFF CHAIN

Files, memory, proofs on Walrus

Index the pointer on Sui. Fetch from Walrus.

PRODUCTION

Public endpoints are for dev

Use a provider or cloud node. Check quotas.

Same lesson today: pick the right read path, stream it, never miss a payment.

Who serves Sui data now.

FULL STACK

gRPC, GraphQL, archive

QuickNode · Ankr · BlockPI
ZAN · Inodra · Triton One

gRPC + GraphQL, no archive:
NodeInfra · 01.ro

gRPC NODES

Full-node gRPC only

Alchemy · Dwellir
BlockVision · GetBlock
Natsai

Pair with GraphQL or an archive
from someone else

INDEXED DATA + STREAMS

Ready-made APIs

Sentio · Surflux (streams)
BlockVision · Blockberry
Space and Time · ZettaBlock
Birdeye · Dubhe Indexer

When you want answers,
not raw checkpoints

Sui Foundation endpoints (fullnode · graphql · archive.mainnet.sui.io): rate limited, for development only.

Before launch, ask: gRPC? GraphQL? Subscriptions over SSE? Archive? Networks? Quotas?

Replace the data path your app depends on.

Start with what the screen or worker needs to do.

APP NEED	REPLACEMENT PATH	TILL EXAMPLE
State + execution	gRPC	Read balances; submit a payment
Filtered activity	GraphQL query + subscription	Load receipts; receive new ones
Pruned history	Archive / backed GraphQL	Find an old transaction by digest

July: Foundation mainnet shutoff → September: GraphQL subscriptions → Migrate each call

A matching transaction becomes a receipt.

```
subscription Till($shop: SuiAddress!,
 $after: String, $afterCheckpoint: UInt53) {
 transactions(filter: { affectedAddress: $shop,
   afterCheckpoint: $afterCheckpoint }, after: $after) {
   cursor
   node { digest sender { address }
     effects { status timestamp
       checkpoint { sequenceNumber }
       balanceChanges(first: 50) { nodes {
         owner { address } coinType { repr } amount
       }}
     }
   }
 }
}
```

CURSOR

Where to resume delivery

DIGEST + STATUS

Transaction + outcome

OWNER + COIN + AMOUNT

Net credit to this shop

TIMESTAMP + CHECKPOINT

When the receipt belongs

SUCCESS + positive net credit
→ receipt → render + chime

One POST request. Many SSE deliveries.

```
const res = await fetch(url, {
 method: 'POST',
 headers: {
   'content-type': 'application/json',
   accept: 'text/event-stream',
 },
 body: JSON.stringify({ query, variables }),
 signal,
});
```

REQUEST

POST /graphql/subscriptions
query + variables

RESPONSE STAYS OPEN

text/event-stream
receipt A → B → C → …

fetch → decode text → buffer complete SSE frames → inspect errors → await the handler

EventSource cannot send this POST body. A network chunk is not a complete SSE message.

The connection drops. The payments do not.



Saved cA

Reconnect: after = cA

B → C → D

No cursor yet? Save the opening checkpoint, then use filter.afterCheckpoint.

Order: checkpoint → transaction position → event position. Persist before advancing.

Lab: prove each part of the payment journey.

ACTION	EXPECTED EVIDENCE	PART OF THE STACK
Open + pay	One receipt + chime; shop credit	gRPC execution + GraphQL feed
Pause → pay → resume	Missed digest appears	Cursor backfill → live
Reload / restart writer	Same receipt counted once	Deduplication + durable cursor
pnpm totals	DB total for the chosen UTC day	SQLite writer + BigInt
Find old receipt	Date + checkpoint + SUCCESS	Archive-backed lookup

TESTNET / VERIFIED PAYMENT PATH: SUI / USDC RUN STILL PENDING

day3-till.vercel.app

Repo: github.com/Nuel-osas/day3-till

Migrate the client, then check the response.

BEFORE / migration/legacy-jsonrpc.ts

```
import { SuiJsonRpcClient,
 getJsonRpcFullnodeUrl
} from '@mysten/sui/jsonRpc';

const client = new SuiJsonRpcClient({
 url: getJsonRpcFullnodeUrl('testnet'),
 network: 'testnet'
});

client.getBalance({ owner: SHOP })
```

AFTER / migration/migrated-grpc.ts

```
import { SuiGrpcClient }
 from '@mysten/sui/grpc';

const client = new SuiGrpcClient({
 network: 'testnet',
 baseUrl:
   'https://fullnode.testnet.sui.io:443'
});

const { balance } =
 await client.getBalance({ owner: SHOP });
```

Check: result shape + total balance + errors + pagination. Obtain fresh API cursors.

Your app: replace one call and prove it works.

01	Find one call	What does its screen need?
02	Choose + replace	gRPC, GraphQL or a subscription
03	Verify the behaviour	Known data + one failure case

Leave with a working diff, its evidence, and a reason for your data-stack choice.

github.com/Nuel-osas/day3-till

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